E4-16
AC6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2012

MSc and EEE PART IV: MEng and ACGI

ANALOGUE SIGNAL PROCESSING

SAMPLE PAPER

Time allowed: 3 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible	First Marker(s):	P. Georgiou
	Second Marker(s):	K. Fobelets

1. a) Compare Voltage and Current mode domains as regards to dynamic range, processing and speed.

[4 marks]

b) The equation for the collector emitter current of a Bipolar Junction Transistor is defined as follows:

$$I_{CE} = I_S \exp\left(\frac{V_{BE}}{U_t}\right)$$

Derive the transconductance of the device and show how comparable performance could be achieved using MOS transistors.

[3 marks]

c) The total noise of a MOS transistor is defined as follows:

$$v_n^2 = \frac{K_w}{I_D^p} \cdot \Delta f + \frac{K_f}{A} \cdot \ln\left(\frac{f_h}{f_l}\right)$$

(i) List the two sources of noise, which contribute to this equation.
(ii) Using this equation, show for M=2 stages of Figure 1.1 that the total output noise is dependent on area, bias current and bandwidth.
(iii) Propose three methods with which the noise of the system Figure 1.1 could be reduced.

[10 marks]

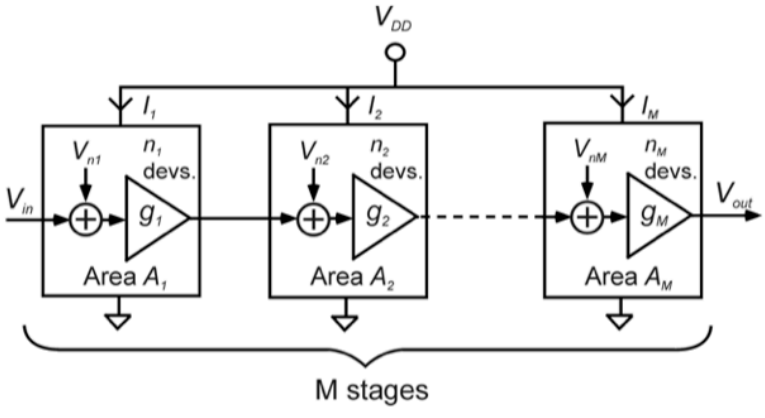


Figure 1.1

2.

a)

(i) What does the term translinear refer to ?

(ii) Draw three devices which could be used as tranlinear elements justifying your choices.

[4 marks]

b) List the three factors need to be satisfied in order to a translinear circuit to work.

[3 marks]

c) Figure 2.1 shows a tranlinear circuit made up of NMOS transistors biased in weak inversion. Derive I_{out} as a function of X by applying the translinear principle. Show all necessay steps. What is the function of the circuit shown in Figure 2.1.

[5 marks]

d) The equation for a two-quadrant mulitplier is is given as:

$$z^+ - z^- = y(x^+ - x^-)$$

Synthesise a translinear circuit to provide this function. You may only use Type A cells and include all reductions where possible.

[8 marks]

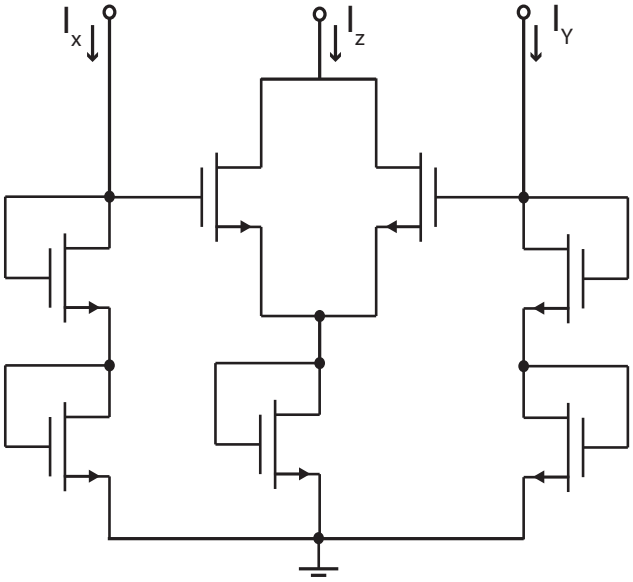


Figure 2.1

3.

a) Figure 3.1 shows a block diagram of a current-mode integrator where f() and g() represent non-linear functions. The input-output relationship of the integrator is linear however:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function $g(Vc)=I_0\exp(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_0, with the function f(x) composed of I_{in}, I_{out} and I_0.

[6 marks]

b) The the transfer function of a low pass filter is defined in state space representation by the following equations:

$$\dot{X} = -\omega_0 X + \omega_0 U$$
$$Y = X$$

whereby Y is the output and U is the inputs and X is the state.

(i) Show that this represents a low pass filter. State any assumptions you make.

[2 marks]

(ii) By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$U = I_U \exp\left(\frac{V_U}{nU_t}\right) \qquad X = I_1 \exp\left(\frac{V_1}{nU_t}\right)$$

[6 marks]

(iii) With these log-domain design equations, sketch a schemtic of the final log domain filter using weak inversion MOS transistors.

[6 marks]

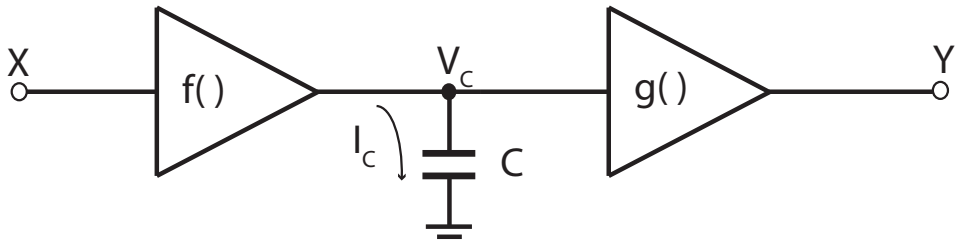


Figure 3.1

4. Figure 4.1 shows a NMOS sampling switch to be used in the switched capacitor circuit. The width and length of transisor M1 are $(W/L)_1=50/2$. This switch causes an error in V_{out} according to:

$$\Delta V = \frac{WLC_{ox}(V_{DD} - V_{in} - V_{TH})}{2C_H}. \qquad (4.1)$$

a)
(i) What is this phenomenon called ?
(ii) Show three other phenomena which could cause an error in V_{out} with equations descibing each case.

[8 marks]

b) Show with the aid of a graph how the error induced by equation 4.1 causes non-ideal gain and offset.

[4 marks]

c) Calculate the total error in the output for Vin=1.65V, CH=10nf , oxide capcitane per unit micron Cox=5 pF /u^2, VDD=3.3V, Vth=0.65V.

[2 marks]

d) Design a method for cancelling out this charge injection stating assumptions and showing calculations where necessary.

[3 marks]

e) What is the advantage of bottom plate sampling of a the capacitor in the circuit Figure 4.2 and state how this should be connected so as not to degrade the circuits speed and precision.

[3 marks]

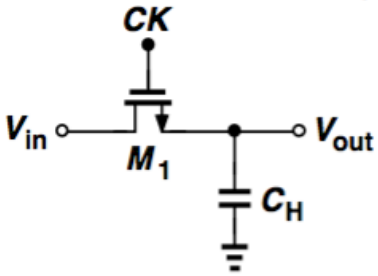


Figure 4.1

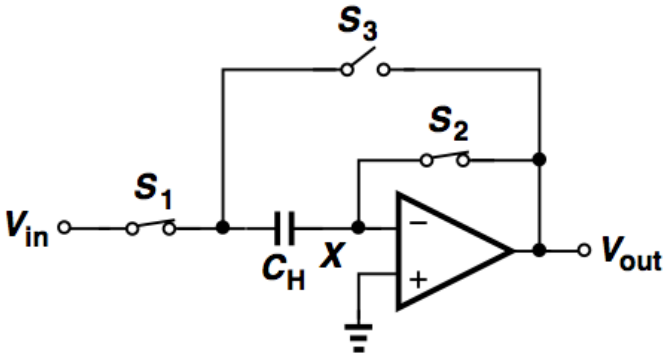


Figure 4.2

5.

a) Explain the principle of autozeroing with the aid of diagrams.
b) Explain how autozeroing affects the frequency spectrum of the signal.

[4 marks]

c) Figure 5.1 shows a closed loop autozeroing amplifier.

(i) Explain how this amplifier removes any offset in the input stage and derive the input referred residual offset , $V_{os\text{-}res}$.

[4 marks]

(i) The Autozeroing amplifier in Figure 5.1 has an offset V_{os}=100mV, a gain A=20,000 and a chrage of $q_{inj}=6x10^{-15}$ C is inserted each time the switch closes. For your derivation in part b.i, calculate the required size of the storage capacitor such that the input reffered offset is always below $V_{os\text{-}res} < 10uV$.

[4 marks]

d) Figure 5.2 shows a switched capacitor amplifier.

(i) Derive the output voltage of this amplifier Vout for both phases Φ1 and Φ2 showing how this removes any offset in the amplifier. You may ignore feedback capacitor C_{dg}.

[4 marks]

(ii) Explain why this configuration requires an opamp with a high slew rate ?
(iii) Show how this circuit can be modified reduce the high slew rate requirements.

[4 marks]

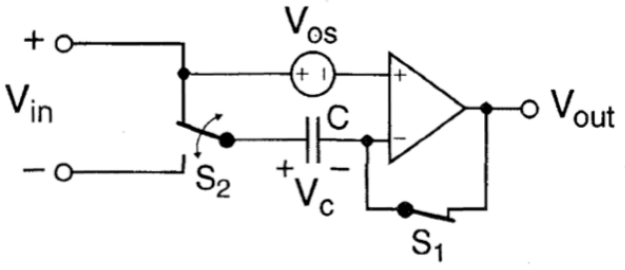


Figure 5.1

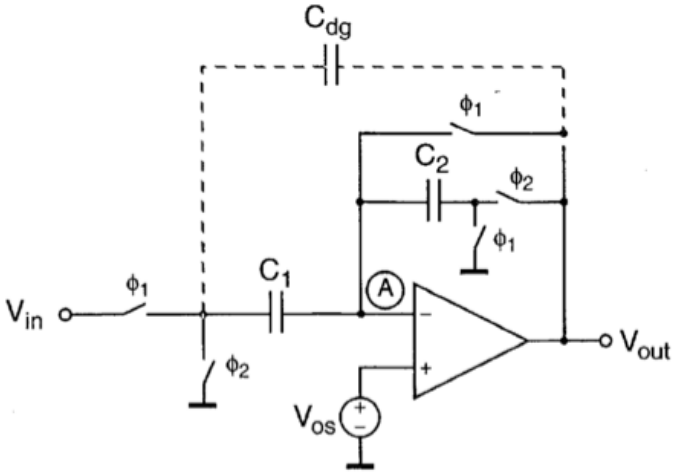


Figure 5.2

6. Figure 6.1 shows a trignonometric circuit whereby all transistors are operating in the weak inversion region of operation with the output currents given by:

$$I_1 = \frac{I_{DC}}{2} e^{\left(\frac{V_+ - V_-}{nU_t}\right)} \qquad I_2 = \frac{I_{DC}}{2} e^{\left(\frac{-(V_+ - V_-)}{nU_t}\right)}$$

a) Show how the circuit can be transformed to give the following trigonometric functions:

$$I_{out1} = I_{DC} \sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_{out2} = I_{DC} \cosh\left(\frac{V_1 - V_2}{nU_t}\right)$$

[6 marks]

b) Figure 4.2 shows a differential pair whereby both transistors M1 and M2 are biased in the weak inversion region of operation. Derive the function for the output current showing how it is trigonometrically related to the input voltage.

[6 marks]

c) Using the two circuits from parts a and b or otherwise design a circuit which implements the following trigonometric function:

$$I_{out} = I_{bias} \operatorname{sech} \frac{V_1 - V_2}{nU_t}$$

whereby I_{bias} is an individually tuned current.

[8 marks]

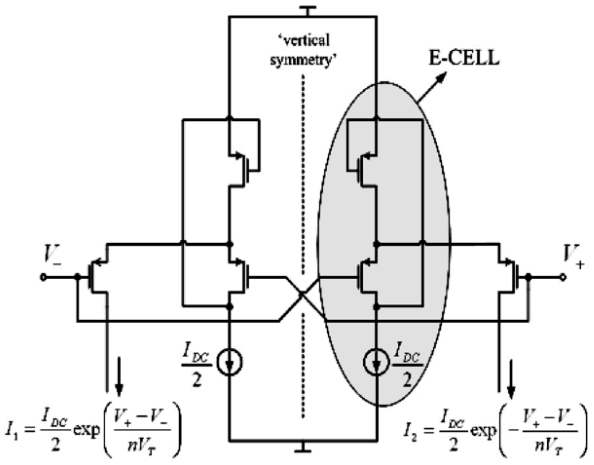


Figure 6.1

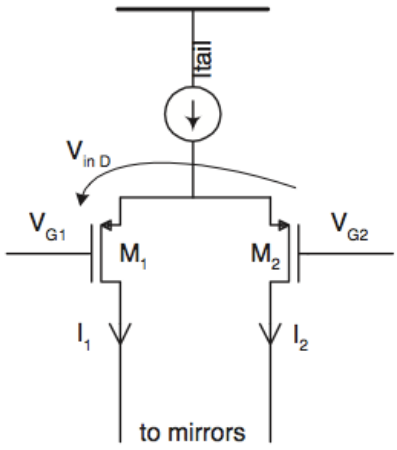


Figure 6.2

2012 ANALOGUE SIGNAL PROCESSING SAMPLE SOLUTIONS

1. a) Voltage mode dynamic range is limited by supply voltages. Current mode dynamic range is limited to how much current can be sourced/sinked leading to larger dynamic range since it is supply independant.

Current mode can lead to easier signal processing since addition of signals requires just summing through Kirchoffs current law.

Current mode operation leads to higher speed of operation since capacitive nodes are already charged.

b) $I_{CE} = I_S \exp\left(\frac{V_{BE}}{U_t}\right)$

Transconductance: $g_m = \frac{dI_{CE}}{dV_{BE}} = \frac{I_S \exp\left(\frac{V_{BE}}{U_t}\right)}{U_t} = \frac{I_{CE}}{U_t}$

MOSFET in subthreshold gives comparable performance!

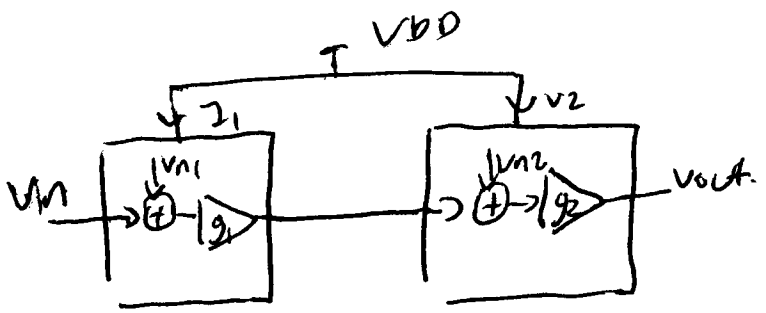
$$I_{DS} = I_0 \exp\left(\frac{V_{GS}}{nU_t}\right)$$

$$g_m = \frac{I_{DS}}{nU_t}.$$

c) $V_n^2 = \frac{K_W}{I_D^{\beta}} \Delta f + \frac{K_f}{A} \ln\left(\frac{f_h}{f_l}\right)$

i) Thermal noise and Vf noise (flicker noise)

ii) Two stages



Distributed gain $G_i = \prod_{K_i}^{K=2} g_K$.

$G_1 = g_1 \cdot g_2$.

$G_2 = g_2$.

Total output noise: $V_{no}^2 = \sum_{i=1}^{i=2} V_{ni}^2 G_i^2$.

$$= \sum_{i=1}^{i=2} \left(n_i \frac{K_W}{(I_i/n_i)^{\beta}} \cdot \Delta f + n_i \frac{K_f}{(A_i/n_i)} \ln\left(\frac{f_h}{f_l}\right) \right) G_i^2$$

Bias current — Area — Bandwidth

iii) To reduce noise of the first stage by increasing area, increasing bias current or operating slowly

2. a) i) Translinear → Trans conductance which is linear with current

ii) Three devices:

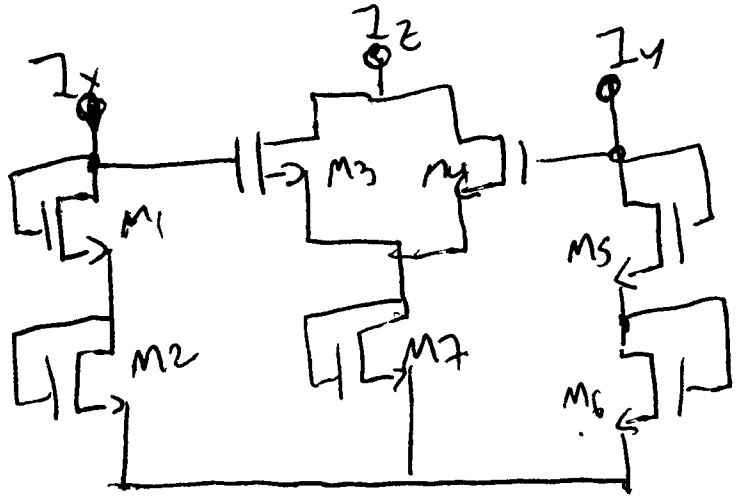
Diode	BJT	MOSFET (WI)
$I_D = I_S \exp(\frac{V_D}{V_t})$	$I_{CE} = I_S \exp(\frac{V_{BE}}{V_t})$	$I_{DS} = I_0 \exp(\frac{V_{GS}}{nU_t})$
$g_m = \frac{I_D}{V_t}$	$g_m = \frac{I_{CE}}{V_t}$	$g_m = \frac{I_{DS}}{nU_t}$

b) 1) Even number of junctions

2) Equal number of clockwise and canter clockwise junctions.

3) Temperatures must be equal.

c)



First loop:
M2 M1 M3 M7.

$I_x^2 = I_3 \cdot I_z$

Second loop
M6, M5, M4, M7.

$I_y^2 = I_4 \cdot I_z$.

$I_4 = I_z - I_3 = I_z - \frac{I_x^2}{I_z}$.

Final result:

$I_z = \sqrt{I_x^2 + I_y^2}$

Vector Sum Circuit

e) $z = xy$, $x = x^+ - x^-$, $z = z^+ - z^-$

Represent as ratios:

$$x^+ = \frac{I_{x^+}}{I_u} \quad , x^- = \frac{I_{x^-}}{I_u} \quad y = \frac{I_y}{I_u} \quad , z^+ = \frac{I_{z^+}}{I_u} \quad , z^- = \frac{I_{z^-}}{I_u}$$

Substitute into

$$(z^+ - z^-) = y(x^+ - x^-)$$

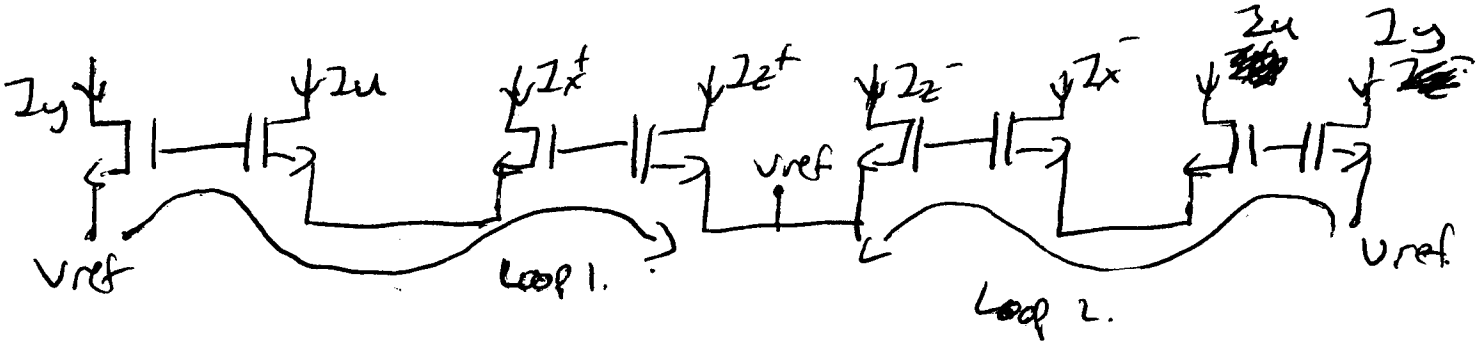
$$\left(\frac{I_{z^+}}{I_u} - \frac{I_{z^-}}{I_u}\right) = \frac{I_y}{I_u}\left(\frac{I_{x^+}}{I_u} - \frac{I_{x^-}}{I_u}\right)$$

re-arranging:

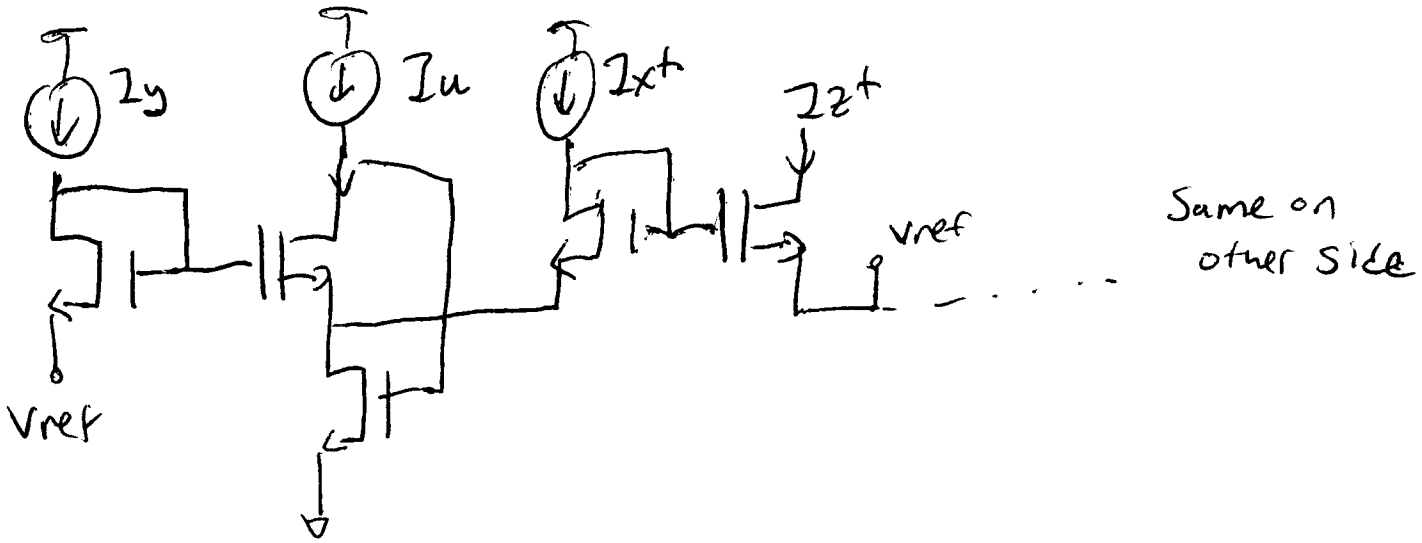
$$I_u \cdot I_{z^+} - I_u I_{z^-} = I_y I_{x^+} - I_y I_{x^-}$$

Synthesise:

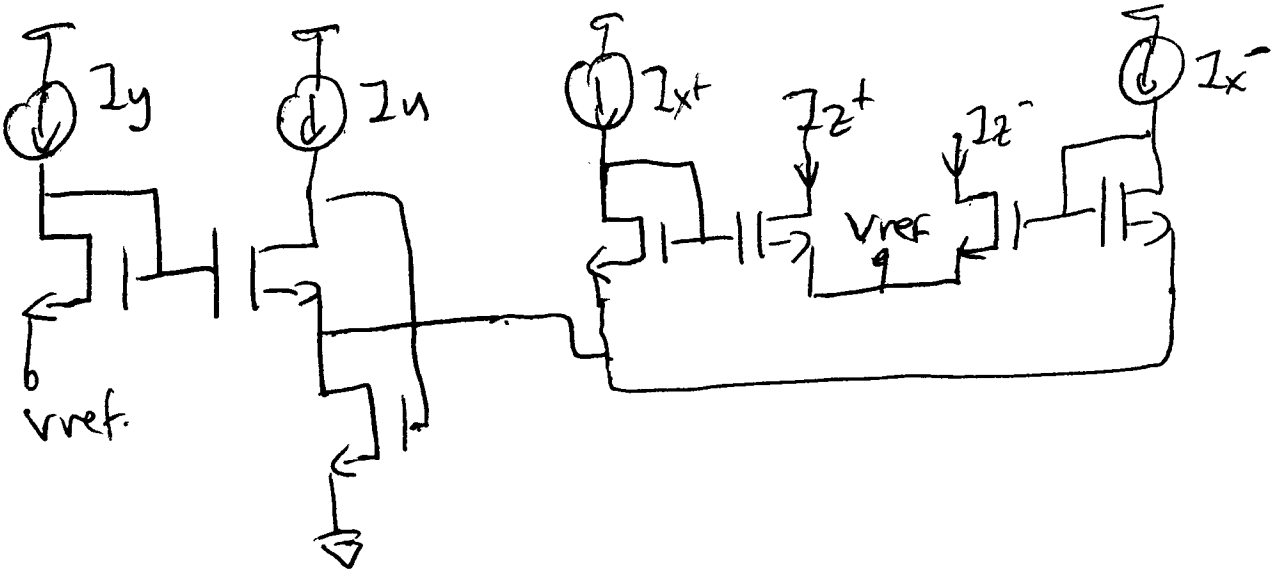
$$\underbrace{I_u \cdot I_{z^+}}_{cw} = \underbrace{I_y I_{x^+}}_{ccw} \quad \text{and} \quad \underbrace{I_u \cdot I_{z^-}}_{cw} = \underbrace{I_y \cdot I_{x^-}}_{ccw}$$



add biasing



Reduce ~~Sig~~ Circuit. by identifying common voltage points:



3. a.)

$$y = g(V_c) = I_0 \exp\left(\frac{V_c}{nU_t}\right) = I_{out}$$

$$\frac{dy}{dV_c} = \frac{dI_{out}}{dV_c} = \frac{I_0}{nU_t} \exp\left(\frac{V_c}{nU_t}\right) = \frac{I_{out}}{nU_t}.$$

$$f(x) = I_c$$

for linear integration we ~~eq~~ require:

$$\frac{dy}{dt} = 2x.$$

$$\frac{dy}{dt} = \frac{d(g(V_c))}{dV_c} \cdot \frac{dV_c}{dt} = \frac{d(g(V_c))}{dV_c} \cdot \frac{I_c}{C}$$

$$= \frac{d(g(V_c))}{dV_c} \cdot \frac{f(x)}{C} = 2x$$

$$\Rightarrow \frac{I_{out}}{nU_t} \cdot \frac{f(x)}{C} = 2x. \quad x = I_{in}$$

$$f(x) = \frac{nU_t \cdot 2 \cdot C \cdot I_{in}}{I_{out}} = \frac{I_0 \cdot I_{in}}{I_{out}}$$

$$I_0 = 2 \cdot C \cdot n \cdot U_t$$

b) $\dot{x} = -\omega_0 x + \omega_0 U$

$Y = X$.

i) Laplace:

$sX = -\omega_0 X + \omega_0 u$.

$X(s+\omega_0) = \omega_0 u$.

$X = \dfrac{\omega_0 u}{s+\omega_0}$ $\quad X = Y$.

$\dfrac{Y}{u} = \dfrac{\omega_0}{s+\omega_0}$ $\Rightarrow$ low pass filter with centre frequency ω_0.

ii) $\dot{x} = -\omega_0 x + \omega_0 u$.

$$\frac{I_1}{nUt} \exp\left(\frac{V_1}{nUt}\right) \cdot \dot{V}_1 = -\omega_0 I_1 \exp\left(\frac{V_1}{nUt}\right) + \omega_0 I_u \exp\left(\frac{V_u}{nUt}\right)$$

therefore.

$$C\dot{V}_1 + C\omega_0 nUt \rightarrow C\omega_0 \frac{I_u nUt}{I_1} \exp\left(\frac{V_u - V_1}{nUt}\right) = 0$$

group constants:

$I_w = C\omega_0 nUt$

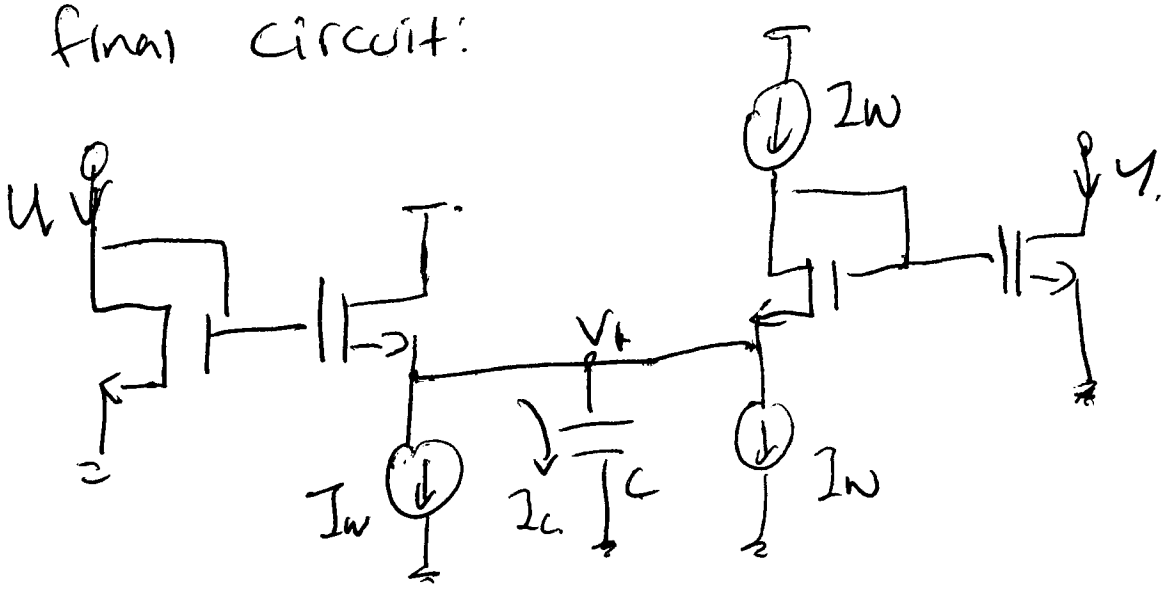
Apply mapping: $I_1 = I_w$

$I_u = I_0$

Final Solution:

$$C\dot{V}_1 = -I_W + I_0 \exp\left(\frac{V_u - V_1}{nU_t}\right)$$

iii) final circuit:



4.

a) Charge injection.

b)

i) Clock feed through:

$$\Delta V = \frac{WC_{ov}}{WC_{ov}+C_H} \cdot V_{ck}.$$

ii) Sampled noise:

$$\Delta V = \sqrt{\frac{kT}{C_H}}$$

iii) ~~Leakage~~ Leakage current:

$$\Delta V = \frac{I_{leak} T_H}{C_H}$$

with descriptions for each.

c) $V_{out} = V_{in} - \Delta V$

$$= V_{in} - \frac{WLC_{ox}(V_{DD} - V_{in} - V_{TH})}{2C_H}$$

$$V_{out} = V_{in}\left(1 + \frac{WLC_{ox}}{2C_H}\right) - \frac{WLC_{ox}}{C_H}(V_{DD}-V_{TH})$$

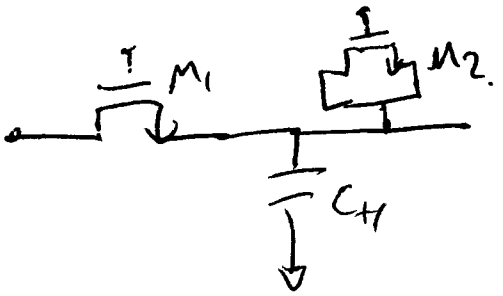
↑ non unity gain

↑ Constant offset.

Correction: 10pf = 10nf for C_H.

d) $\Delta V_{out} = \frac{(50 \times 2 \times 5\,p)(3.3 - 1.65 - 0.65)}{2 + 10\,p \times 10^{-3}} = \frac{50 \times 2 \times 5}{2 \times 10 \times 10^{3}} = 25\,mV.$

e) Cancel out by means of a dummy switch.

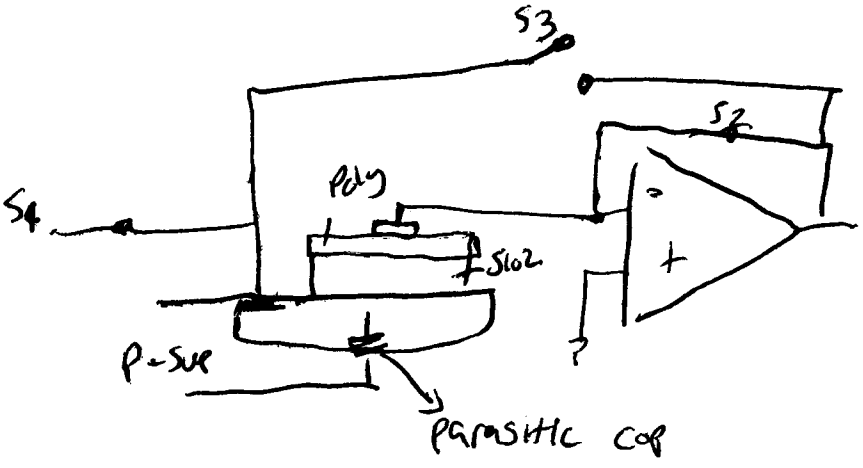


Choose $W_2 = 0.5 W_1$, $L = L_2$ to ensure $q_1 = q_2$.

$W_2 = 25u \quad L_2 = 2u$

f) . Input capacitance of opamp degrades speed and precision.

. Bottom plate of capacitor contains a parasitic capacitance to substrate and should therefore not be connected to input of opamp

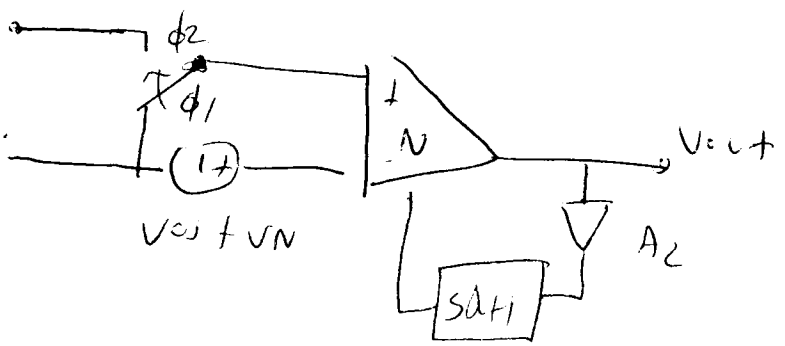


. avoids injection of substrate noise

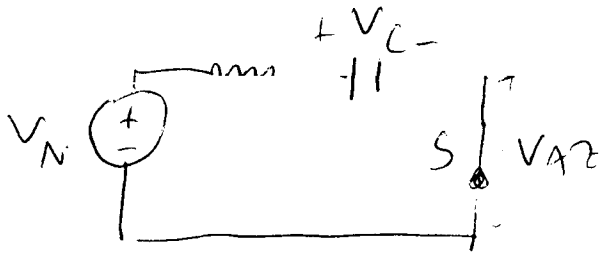
5. Auto-zeroing is a method of reducing offset and low frequency noise commonly found in amplifiers.

Basic Principle.

- Sample the unwanted quantity (noise + offset)
- Subtract this from the instantaneous value of the contaminated signal.
- The cancellation can occur at the input or an intermediate node between input and output



- Constant noise gets subtracted.
- Low frequency 1/f noise will be high pass filtered and thus reduced at low frequencies. Wide band noise will be aliased.



switch closed

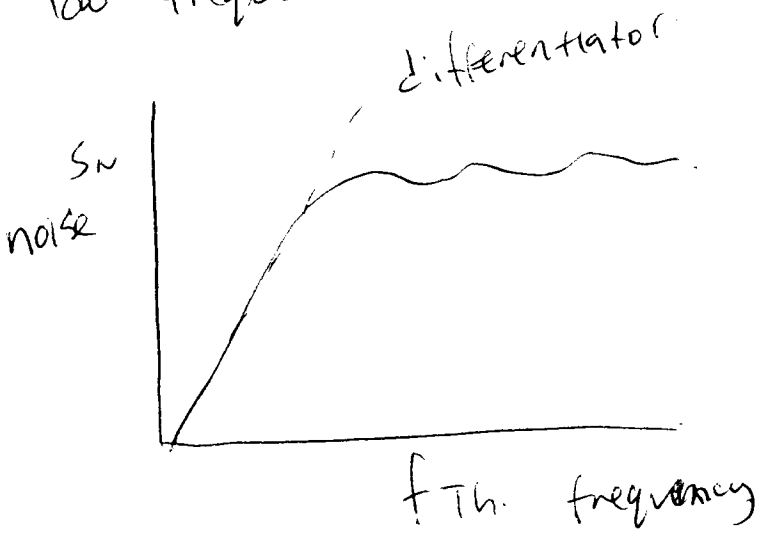
$V_C = V_N = V_{AZ}$

Switch open

$V_{AZ} = V_N - V_C = 0$!

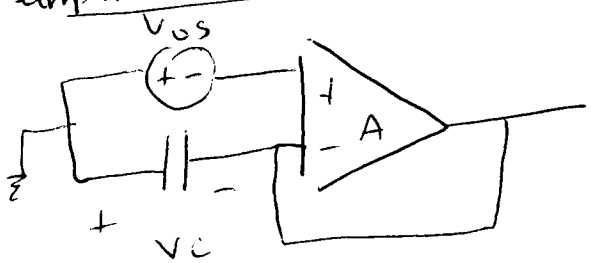
b) Slow moving 1/f noise is like DC at signal frequencies. It will get subtracted.

Output is a differentiated version of the input at low frequencies.



C. Offset is sampled on ~~Vc then Subtra~~ Capacitor C then subtracted.

ϕ_1 sampling phase



$V_{out} = A(-V_{os} + V_c)$

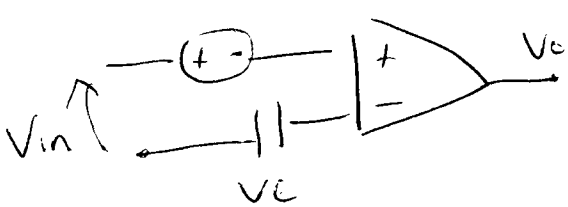
$V_{out} = -V_c$.

$-V_c = -AV_{os} + AV_c$.

$AV_{os} = (1+A)V_c$.

$V_c = \frac{A}{1+A} V_{os}$

ϕ_2. Subtracting phase



$V_c = V_{os} \cdot \frac{A}{1+A} + \frac{q_{in}}{C}$.

$V_{out} = A(V_{os} - V_c)$

$= A\left(V_{os} - \frac{AV_{os}}{1+A} + \frac{q_{in}}{C}\right)$

$\frac{V_{out}}{A} = \frac{V_{os}}{1+A} + \frac{q_{in}}{C} = V_{os_res}$

ii) Corrections: $A = 20{,}000$, $q_{inj} = 6\times10^{-15}$ C.

$$V_{os-res} = \frac{V_{os}}{A} + \frac{q_{inj}}{C} = \frac{100\times10^{-3}}{20\times10^{3}} + \frac{6\times10^{-15}}{C} = 10\mu V$$

$$C = \frac{6\times10^{-15}}{10\mu - 5\times10^{-6}} = 1.2nf.$$

d) Φ_1:

i) $$Q_1' = C_1(V_{in} - V_{os})$$
$$Q_2' = C_2 V_{os}.$$

Φ_2:
$$Q_1 = C_1(-V_{os})$$
$$Q_2 = C_2(V_{os} - V_{out}).$$

Charge conservation on node A means:

$$Q_1' - Q_1 = Q_2' - C_2$$

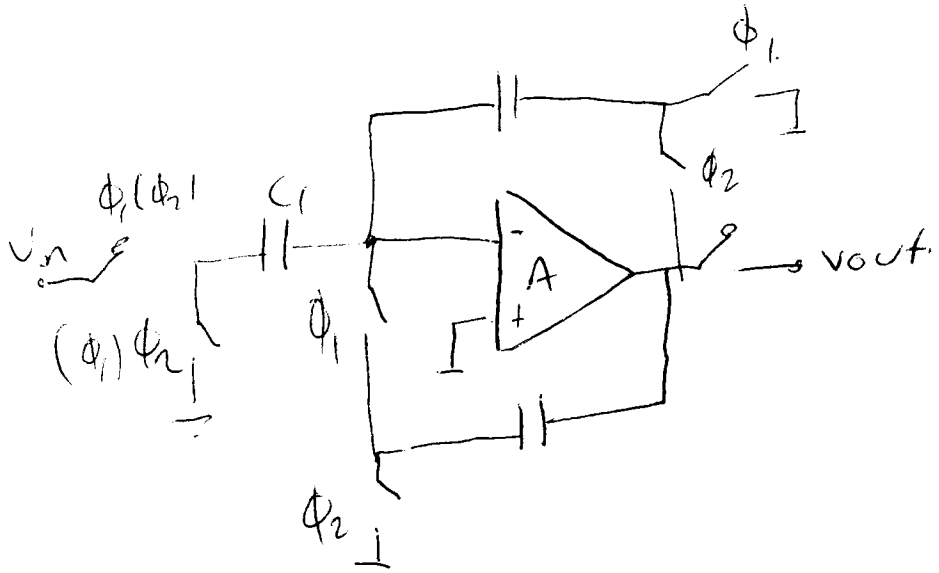
Solving:

Φ_2 $V_{out} = -V_{in}\frac{C_1}{C_2}$; independant of V_{os}

Φ_1 $V_{out} = V_{os}$.

ii) Output V_{out} has to change from $-V_{in}\frac{C_1}{C_2}$ to V_{os} during Sampling Phase Φ_1 and Amplification Phase Φ_2. This requires high Slew rate to accomodate this Swing.

iii) Improved version which doesn't require high slew rate:



Parenthesis : inverting amplifier

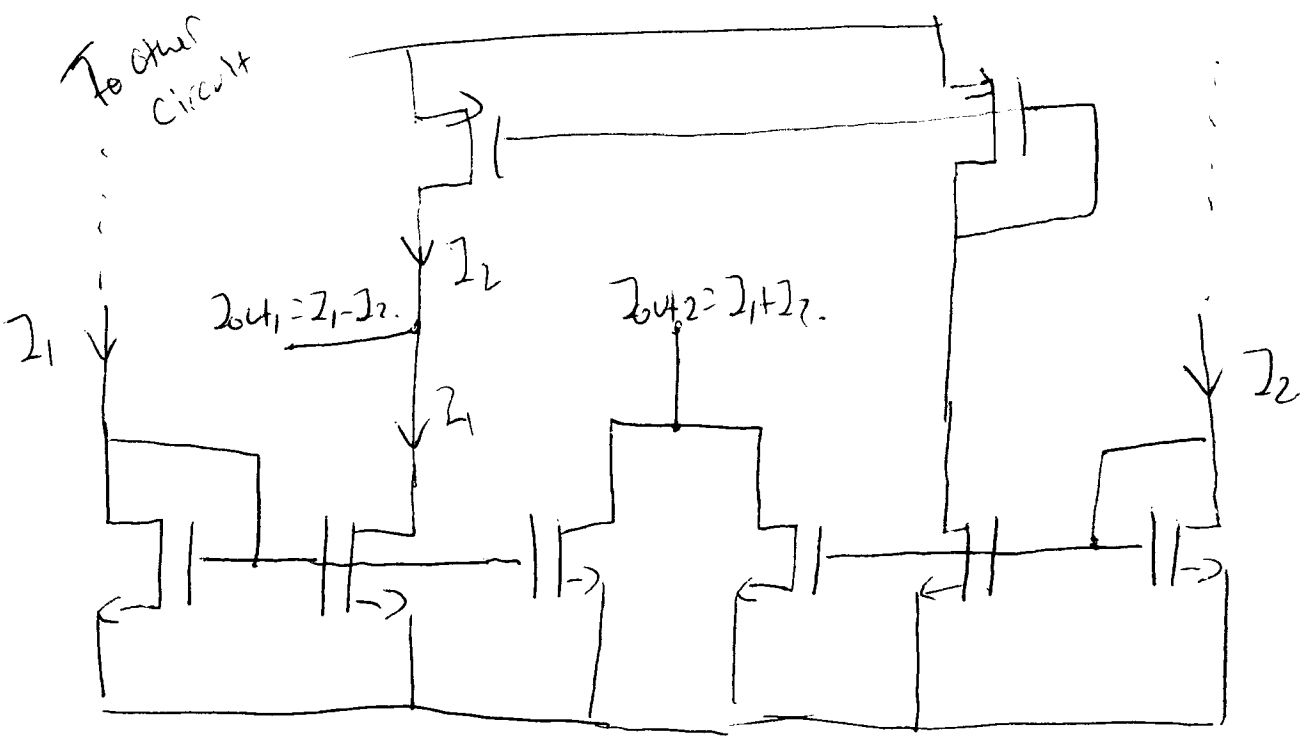
out side Parenthesis : non-inverting amplifier.

6. $\sinh(x) = \frac{e^x - e^{-x}}{2}$

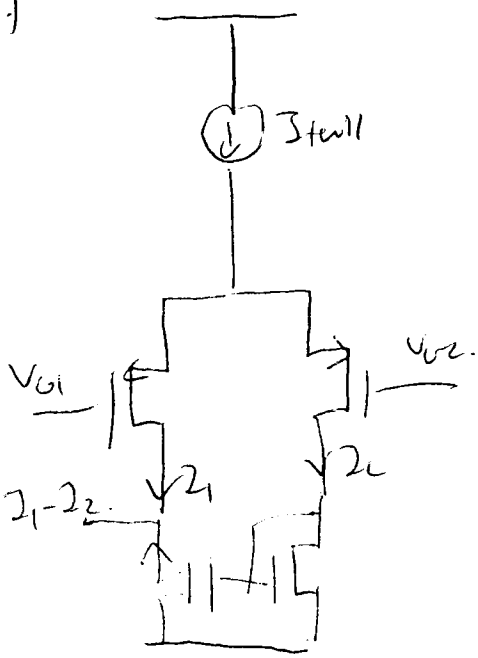
$\cosh(x) = \frac{e^x + e^{-x}}{2}$.

$I_1 - I_2 = I_{DC} \sinh\left(\frac{V_1 - V_2}{nU_t}\right)$

$I_1 + I_2 = I_{DC} \cosh\left(\frac{V_1 - V_2}{nU_t}\right)$



b)



$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

$$I_1 = I_0 \exp\left(\frac{V_{GS1}}{nU_t}\right)$$

$$I_2 = I_0 \exp\left(\frac{V_{GS2}}{nU_t}\right)$$

$$I_{tail} = I_1 + I_2 = I_0\left(e^{\frac{V_{GS1}}{nU_t}} + e^{\frac{V_{GS2}}{nU_t}}\right)$$

$$I_{out} = I_1 - I_2 = I_0\left(e^{V_{GS1}/nU_t} - e^{V_{GS2}/nU_t}\right)$$

$$\frac{I_{out}}{I_{tail}} = \frac{e^{V_{GS1}/nU_t} - e^{V_{GS2}/nU_t}}{e^{V_{GS1}/nU_t} + e^{V_{GS2}/nU_t}} = \tanh\left(\frac{V_{in}}{nU_t}\right)$$

$$V_{in} = V_{GS1} - V_{GS2}$$

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c) Combine two circuits:

$$I_{out1} = I_{0C} \sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_{out2} = I_\alpha \tanh\left(\frac{V_1 - V_2}{nU_t}\right) = I_\alpha \frac{\sinh\left(V_1 - V_2/nU_t\right)}{\cosh\left(V_1 - V_2/nU_t\right)}$$

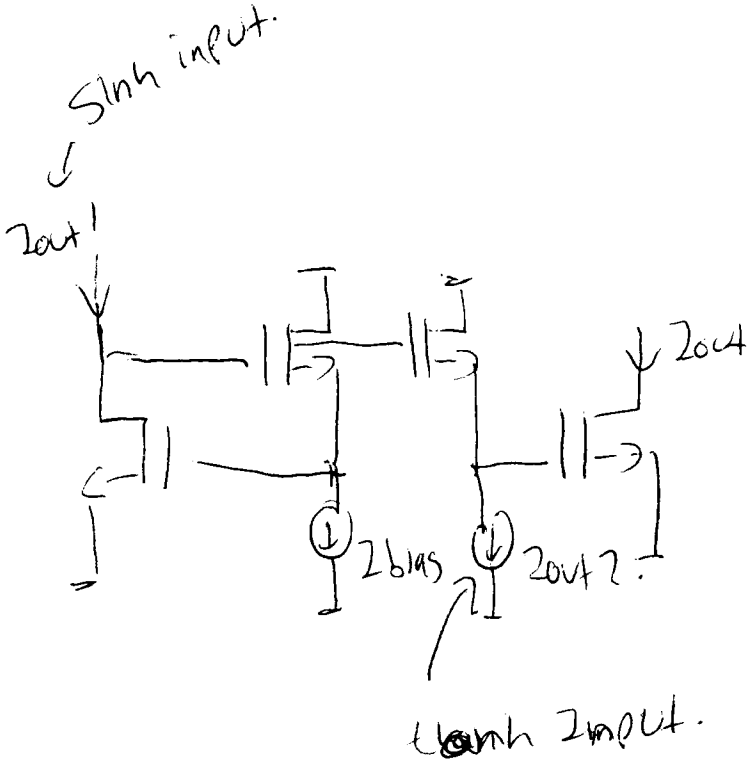
$$\frac{I_{out1}}{I_{out2}} = \frac{1}{\cosh\left(\frac{V_1 - V_2}{nU_t}\right)} = \operatorname{sech}\left(\frac{V_1 - V_2}{nU_t}\right)$$

multiply by I_{bias}:

$$I_{bias}\frac{I_{out1}}{I_{out2}} = I_{bias}\operatorname{sech}\left(\frac{V_1 - V_2}{nU_t}\right)$$

translinear cell! →

$$I_{OUT} = I_{bias} \cdot \frac{I_{out1}}{I_{out2}} = I_{bias} \tanh\left(\frac{V_1 - V_2}{2V_t}\right)$$



- Draw complete circuit!